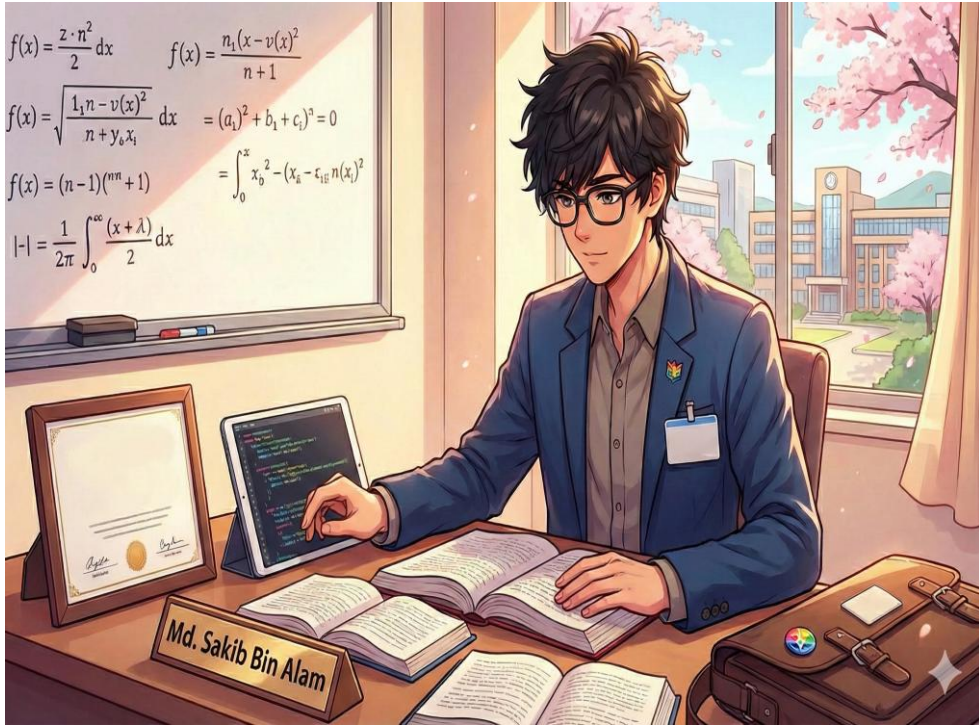




Engineering Beyond Borders: Global Higher Education & Career Pathways

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Experience

Assistant Professor (on leave), UITS



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Understanding Graduate Programs

Navigating the Path to a Master's or PhD



Master's / PhD

Master's

Master's is an advanced academic degree focusing on specialized knowledge and skills.

More of a professional or specialized training that makes you more career-focused.

PhD

PhD is a research-intensive degree aimed at original contributions to a field.

A more research-oriented degree that focuses on creating new dimensions of knowledge

Master's

Usually **2 years to complete** / **Specialized training** / **Industry based** / Tuition/**Stipend**/Insurance:
Rarely Provided

Masters by Coursework

Structured Learning / Career Oriented / Skill Acquisition Funding: Usually Self-Funded (though scholarships exist)

Masters by Thesis (Masters by Research / MPhil)

Independent Study / Discovery Oriented / Academic Rigor Funding: Stipends/Tuition Offsets often available

PhD



Usually **3-5 years to complete** / **Research Oriented** / Creating **new dimensions of Knowledge**
Tuition/**Stipend**/Insurance: Usually Provided

Best for Research Career

Requires a **long-term commitment**.

Adheres to **institution-defined rigorous thesis standards**.

Best suited for **careers in academia and research**.

The success of a PhD hinges on a strong supervisor and a compelling research topic.

A **PhD program** typically follows this structure:

- ✓ May have Coursework (1-2 years)
- ✓ Qualifying/Comprehensive Exams
- ✓ Research Proposal
- ✓ Dissertation Research (3-5 years)
- ✓ Publication & Conferences
- ✓ Dissertation Writing & Defense

PhD Research

Research is a systematic process of **investigating, analyzing, and generating new knowledge** or solutions to existing problems through structured inquiry and critical thinking.

Select Your Research, Responsibly

Do's:

- ✓ Choose an area that fascinates you.
- ✓ Leverage your existing experience or knowledge.
- ✓ Ensure you have a strong understanding of the topic.
- ✓ Select something that aligns with your strengths.
- ✓ Consider areas with strong industry demand.

Don'ts:

- ✗ Choose it merely because it's trendy or in high demand in the job market.
- ✗ Pursue it just for the sake of studying abroad.
- ✗ Select it when you feel you don't have a clear understanding of the subject.

Building your profile
requires time and effort

The Journey is challenging

No instant return on
investments

Being away from family
can be difficult

Adopting a new lifestyle

Financial rewards take
time

Funding Options

Scholarships, Assistantships, Financial
Planning for Graduate Programs



Funding Options

PhD students often receive **full funding (RA, TA, fellowships)**, while master's students have **few funded opportunities** and may rely on scholarships or self-funding.

Funded Opportunity



- 👉 Offers a stipend and tuition exemption in return for teaching responsibilities.
- 👉 Supported by faculty research funds, requiring assistance in research activities.
- 👉 Involves academic or administrative support tasks, typically with partial tuition assistance.
- 👉 Competitive, merit-based funding that covers tuition and stipend without work obligations.

Self-Funded



- 👉 Self-funded studies or financial support from employers or sponsors.

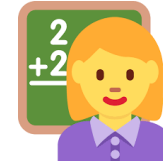
Funded Opportunities



Research Assistant

- Conduct literature reviews, experiments, and data analysis for faculty research projects.
- Assist in writing research papers, reports, and grant proposals.
- Maintain lab equipment, manage datasets, and support research-related tasks.

Teaching Assistant



- Conduct tutorials, lab sessions, or discussion groups to support student learning.
- Grade assignments, exams, and provide feedback to students.
- Assist professors with course preparation, materials, and administrative tasks.

Funded Opportunities (Others)

Fellowship and Scholarships



Example: GU Postgraduate Research Scholarship

Country Specific Scholarships



Example: MEXT (Japan), Govt. Scholarships (China/Turkey), DAAD (Germany), Erasmus

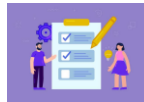
Preparation: **What you need**

Scores



- 👉 Academic Transcripts & CGPA
- 👉 English Proficiency Test (IELTS/TOEFL)
- 👉 GRE/GMAT

Things you collect



- 👉 Letter of Recommendation
- 👉 Financial Documents (self fund)

Things you prepare



- 👉 CV/Resume
- 👉 Statement of Purpose
- 👉 Research Portfolio

Academic Scores



CGPA

CGPA plays a crucial role in higher studies as it reflects academic performance and directly impacts admissions, funding, and scholarship opportunities.

A higher CGPA enhances your chances of securing funding and other academic benefits.

- Requirements vary by university: **some ask for a minimum 3.00 CGPA, others look for the CGPA of the last 2/4 semesters, typically requiring around 3.50.**
- Always check the specific CGPA requirements for each university before applying.

Academic Scores

Evaluation

Assessment of your academic history, including grades and coursework, to determine equivalency to the host institution's standards.

It is generally **not a primary focus** and should not be overly stressed. Most universities **appropriately consider academic transcripts from Bangladeshi Universities**.

However, some universities may require additional evaluations, such as **WES, SpanTran, or ECE**. Therefore, it is important to review the specific university requirements before proceeding with your academic transcript evaluation, as they will usually provide clear guidance on whether an evaluation is needed and which type to use.



WORLD EDUCATION SERVICES



Letter of Recommendation

A Letter of Recommendation (LOR) is a formal document written by a professor, employer, or mentor that evaluates an applicant's qualifications, skills, and potential for higher studies or employment.

- ✓ Typically, **two/three Letters of Recommendation** are required.
- ✓ If you have work experience, **an employer can also provide a recommendation, but academic LORs are generally preferred.**

Reach out to your professor before submitting your application.

LORs are typically submitted through forms that your professor needs to complete carefully.

Along with the form, some universities require an official document where the professor writes a signed letter and uploads it.

Letter of Recommendation

Suggestions:

- 👉 Each LOR should highlight distinct qualities and insights from the professor.
- 👉 Avoid simplistic statements like “he is good”.
- 👉 Keep the letter concise and limited to one page.



Key Content to Include:

- 👉 How does the professor know you?
- 👉 Academic performance in class.
- 👉 Contributions to thesis or research work.
- 👉 Leadership qualities or significant achievements.



CV/Resume



A resume/CV is a concise document summarizing your education, skills, work experience, and achievements, tailored for academic or professional opportunities.

Structure

Header – Include your name and contact details.

About Section – A summary of your goals and aspirations.

Work Experience – Highlight relevant jobs or internships.

Education – List your degrees, institutions, and graduation years.

Publications (if any) – Mention journal articles, conference papers, or books.

Research Experience – Summarize your academic focus and research projects.

Projects – Outline key university or independent projects.

Scholarships & Awards – List notable recognitions and achievements.

Extracurricular Activities – Highlight involvement beyond academics.

Limit your CV to a maximum of two pages.

Provide accurate information; avoid false claims.

An academic CV does not require a photo.

Maintain a consistent font style and size throughout.

Statement of Purpose

A Statement of Purpose (SOP) is a written document outlining an applicant's academic background, career goals, research interests, and reasons for applying to a specific program, helping universities assess their suitability for admission.

Structure

- 👉 Begin with a paragraph outlining your aspirations and objectives.
- 👉 Provide a concise overview of your academic background, professional experiences, and key achievements.
- 👉 Explain your motivation for pursuing higher studies.
- 👉 Highlight your preparation for advanced studies, including ongoing research.
- 👉 Briefly summarize any previous research projects.
- 👉 Justify your choice of university, department, research center/lab, and professor.
- 👉 Describe your intended research focus and contributions.
- 👉 Conclude by explaining why the university should consider you as a candidate.

Statement of Purpose



A standard Statement of Purpose (SOP) typically adheres to the following formatting guidelines:

Font: Use a professional and readable font such as Times New Roman, Arial, or Calibri.

Font Size: Set the font size to 11 or 12 points.

Line Spacing: Employ 1.5 or double spacing to enhance readability.

Margins: Maintain 1-inch (2.5 cm) margins on all sides.

Length: Aim for a length between 500 and 1,000 words, typically fitting within one to two pages.

Universities may have specific guidelines for writing an SOP. Some universities may set specific questions that must be addressed in your SOP.

The key objective is to create a seamless narrative by linking all your past experiences in a logical and coherent manner.

Take your time—perfecting your final SOP draft may take 1-2 months.

Avoid copying or paraphrasing SOPs; while you can seek inspiration, it is best to write your own, as SOPs may be checked for plagiarism.

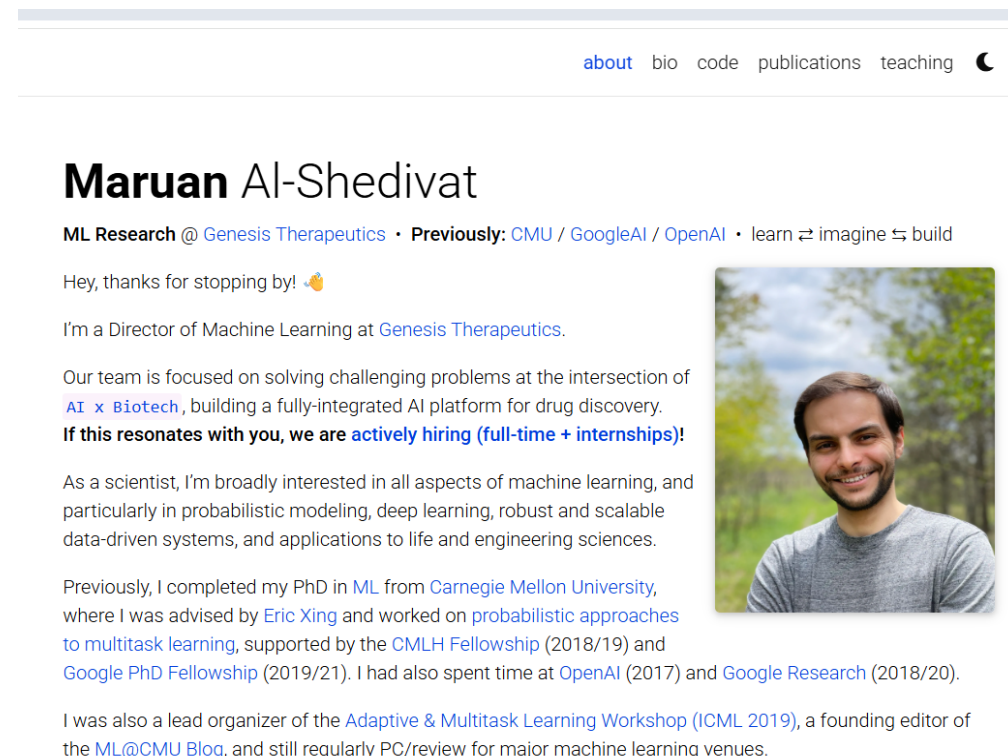
Research Portfolio/Website

A research portfolio is a collection of your academic research work, including publications, projects, presentations, and relevant experiences, showcasing your expertise and contributions in a specific field.

Components:

- Research Statement
- Education
- Work Experience (if any)
- Research Interest
- Publications
- Projects
- Awards

You can use Google Sites,
Static Website w/Github



University Application Journey

Search & Select University

Explore university options



Search & Select Professor & Lab

Identify potential mentors



Wait for Response

Anticipate professor's reply



Email your selected Professor

Initiate contact



Assessment/Interview/Further Communication

Engage with professor



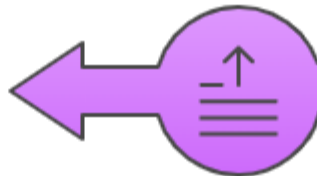
Apply to the University

Submit application



Wait for the Result

Await university decision



Search & Select University

Few Key Factors (to watch out)

- 👉 You can refer to various reputable rankings, such as **USNews** (best for US schools), **Times Higher Education**, and **QS rankings**.
- 👉 However, rankings can be misleading, so it's better to focus on the **ranking of your specific major rather than the overall ranking**.
- 👉 Keep in mind that rankings alone don't tell the whole story—professors, research labs, and personal fit are the most important factors.

CSRankings: Computer Science Rankings

CSRankings is a metrics-based ranking of top computer science institutions based on faculty publications at selective conferences.

Australia 2014 2025 All AI Systems Theory Interdisc. Tour Sponsor CSRankings

All Areas

AI

- Artificial intelligence ☒
- Computer vision ☐
- Machine learning ☐
- Natural language processing ☐
- The Web & information retrieval ☐

Systems

- Computer architecture ☐
- Computer networks ☐
- Computer security ☐
- Databases ☐
- Design automation ☐
- Embedded & real-time systems ☐
- High-performance computing ☐
- Mobile computing ☐
- Measurement & perf. analysis ☐
- Operating systems ☐
- Programming languages ☐

Institution	Count	Faculty
1 ▶ UNSW	52.5	23
2 ▶ Monash University	45.1	36
3 ▶ University of Melbourne	37.2	29
4 ▶ University of Technology Sydney	35.2	19
5 ▶ Macquarie University	34.0	18
6 ▶ University of Sydney	32.4	16
7 ▶ Australian National University	32.1	20
8 ▶ Griffith University	27.1	13
9 ▶ Adelaide University	25.3	18
10 ▶ RMIT University	19.4	22
11 ▶ University of Queensland	6.7	6
12 ▶ University of Wollongong	4.5	7
13 ▶ Western Sydney University	4.2	4
14 ▶ University of Western Australia	3.8	5
15 ▶ Queensland Univ. of Technology	1.7	2

Search & Select University



Weather

Weather can influence university selection, as some students prefer specific climates, which can affect their overall college experience.

Melbourne (Victoria): Famous for its "four seasons in one day," bringing unpredictable changes and cold, damp winters that can be challenging for those used to warmer climates.

North Queensland (e.g., Townsville/Cairns): Experiences intense humidity and heavy monsoon rains during the wet season, with the potential for tropical cyclones.



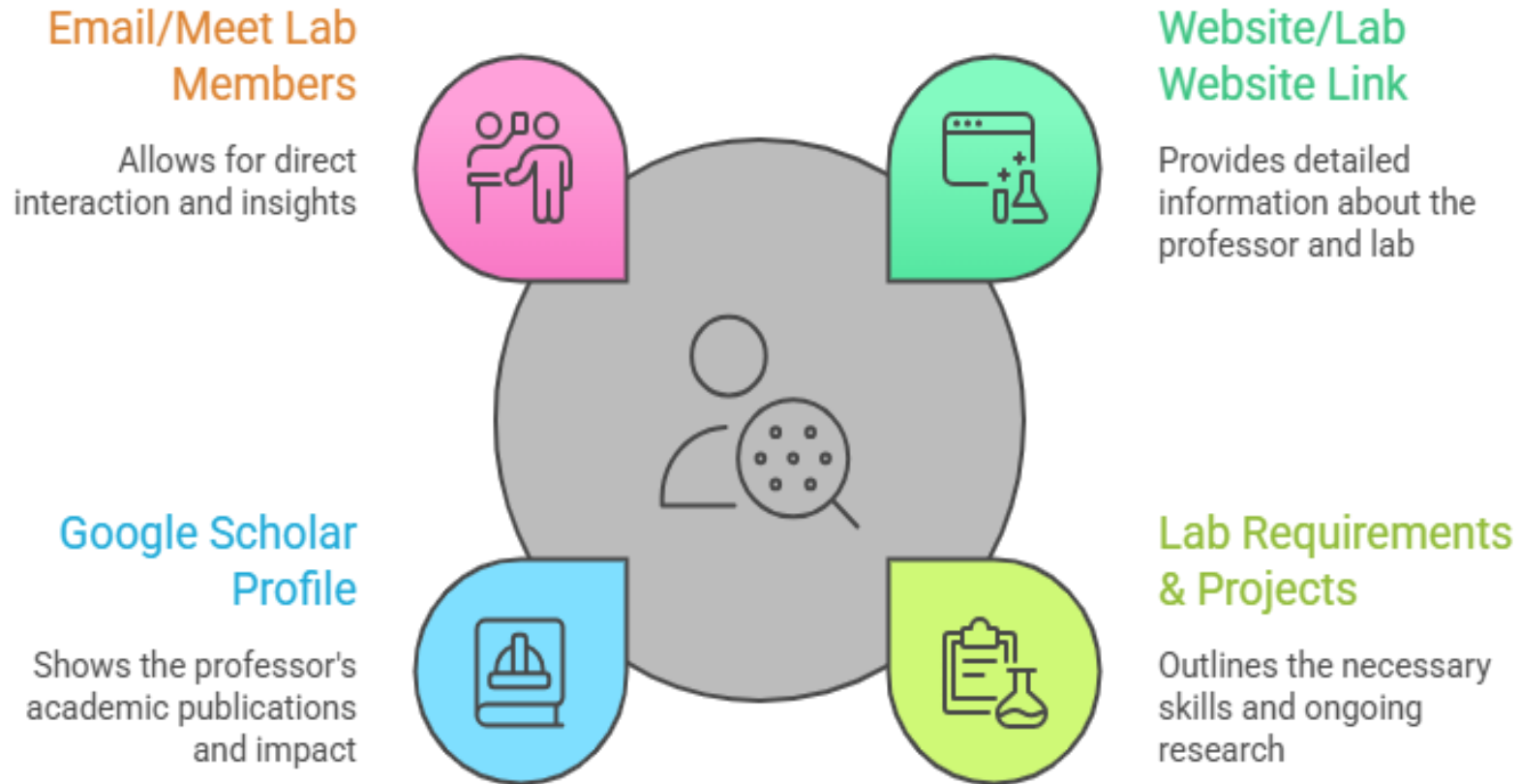
Living Cost

Living costs can significantly influence university selection, as they impact overall affordability and the financial burden on students.

Sydney (New South Wales): Consistently ranked as the most expensive city in Australia, known for exorbitant rental markets and high general cost of living.

Melbourne (Victoria): Features elevated costs in inner-city housing and lifestyle expenses, though slightly more affordable than Sydney.

Finding the Right Professor/Lab



Made with  Napkin

Putting it All Together

Create a Google/Excel sheet to track the following details for the universities you've selected:

University Name, Rank, Priority (based on your preferences), Application Link, Deadline, Decision Time, Department Link, Professor Link, Requirements, and Comments.

This will help you stay organized and plan your journey effectively.

Communication Stage

Now it's time to send emails to Professor. This is the most tedious part, requiring patience and consistency

Collect the e-mail addresses from

- ✓ University Website
- ✓ Faculty Website
- ✓ Opening Advertisements

Patience

Consistency

Communication Stage

Things to Understand before you send an e-mail

- If you're unsure about a professor's research, avoid emailing them. Don't be desperate—focus on professors whose work you truly understand.
- Avoid sending generic emails to multiple professors. Instead, tailor your emails to each one.
- Before reaching out, thoroughly explore the professor's website or scholar profile, read 2-3 of their recent papers, and base your email on that understanding. Be ready to engage with their research by reading the papers and project documents.

Communication Stage

Some E-Mail Suggestions

Time to E-Mail

- US/AUS Time: **8AM – 1PM**
- Sending emails early is ideal, as they will appear at the top of the inbox.

Follow-up Email

- **Follow-up after one week.**
- Second Follow-up after one week of follow-up.

Whom to Avoid

- **Adjunct Professor + Emeritus Professor**
- They usually don't have any funds.

Mailing the Same Department

- If you've selected three professors from the same department, space out your emails by 2-3 days.
- For those within the same research lab, you can contact them simultaneously.

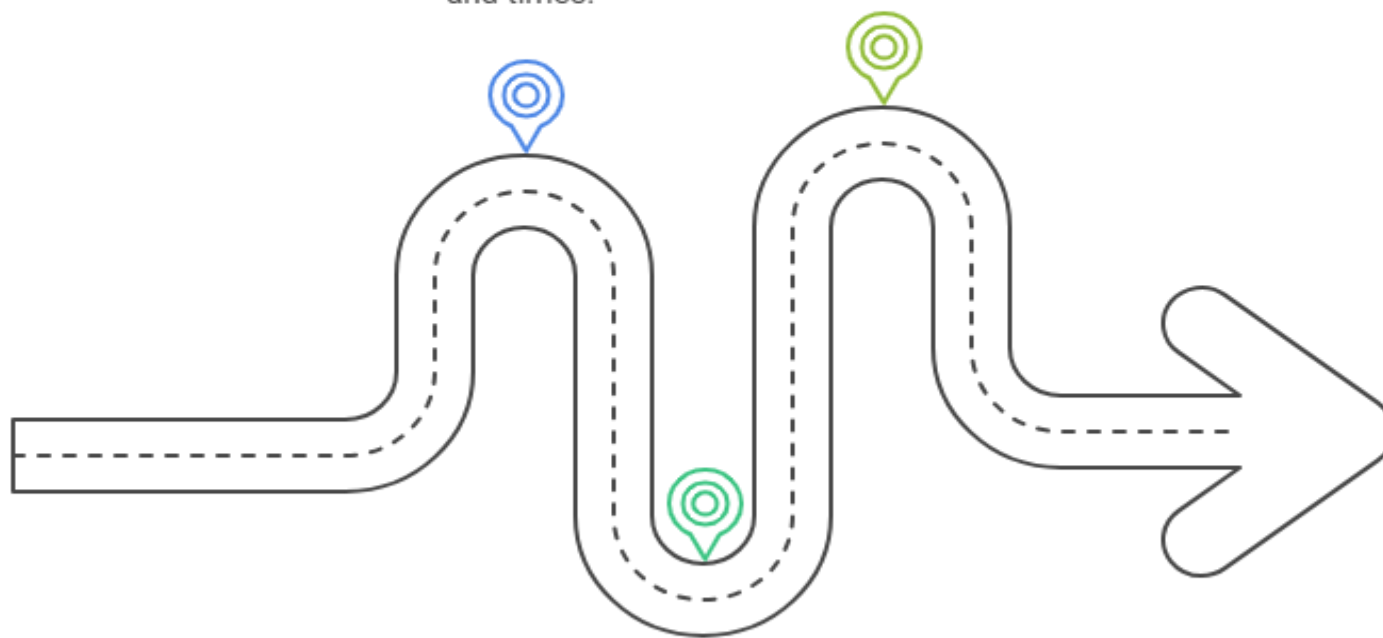
Completing Your Application

Review Deadlines

Understand all
submission dates
and times.

Submit Application

Complete and send
your application
before the deadline.



Check Requirements

Ensure you meet all
necessary criteria.

Career Pathways

Temporary Graduate Visa – Subclass 485(**Post-Higher Education Work Stream**).This visa allows full-time work and is the primary job-search visa after graduation. **Typical Duration: 2-3 years.**

Long-Term Migration Pathways for CS Graduates

Computer Science occupations are highly demanded in Australia's Skilled Occupation Lists.

Common Visa Pathways:

- 👉 Subclass 189 – Skilled Independent (PR)
- 👉 Subclass 190 – State Nominated (PR)
- 👉 Subclass 491 – Skilled Work Regional (Provisional → PR)
- 👉 Subclass 186 / 482 – Employer Sponsored



Career Options After a CS Degree

Industry & Corporate Careers

Most CS graduates move directly into industry roles such as:

- Software Engineer / Developer
- Data Scientist / Data Analyst
- AI / Machine Learning Engineer
- Cybersecurity Analyst
- Cloud Engineer / DevOps Engineer
- Web & Mobile App Developer
- IT Consultant / Systems Analyst
- Game Developer
- Product Engineer / Technical Product Manager

These roles exist across tech companies, banks, consulting firms, healthcare, defence, mining, and startups.

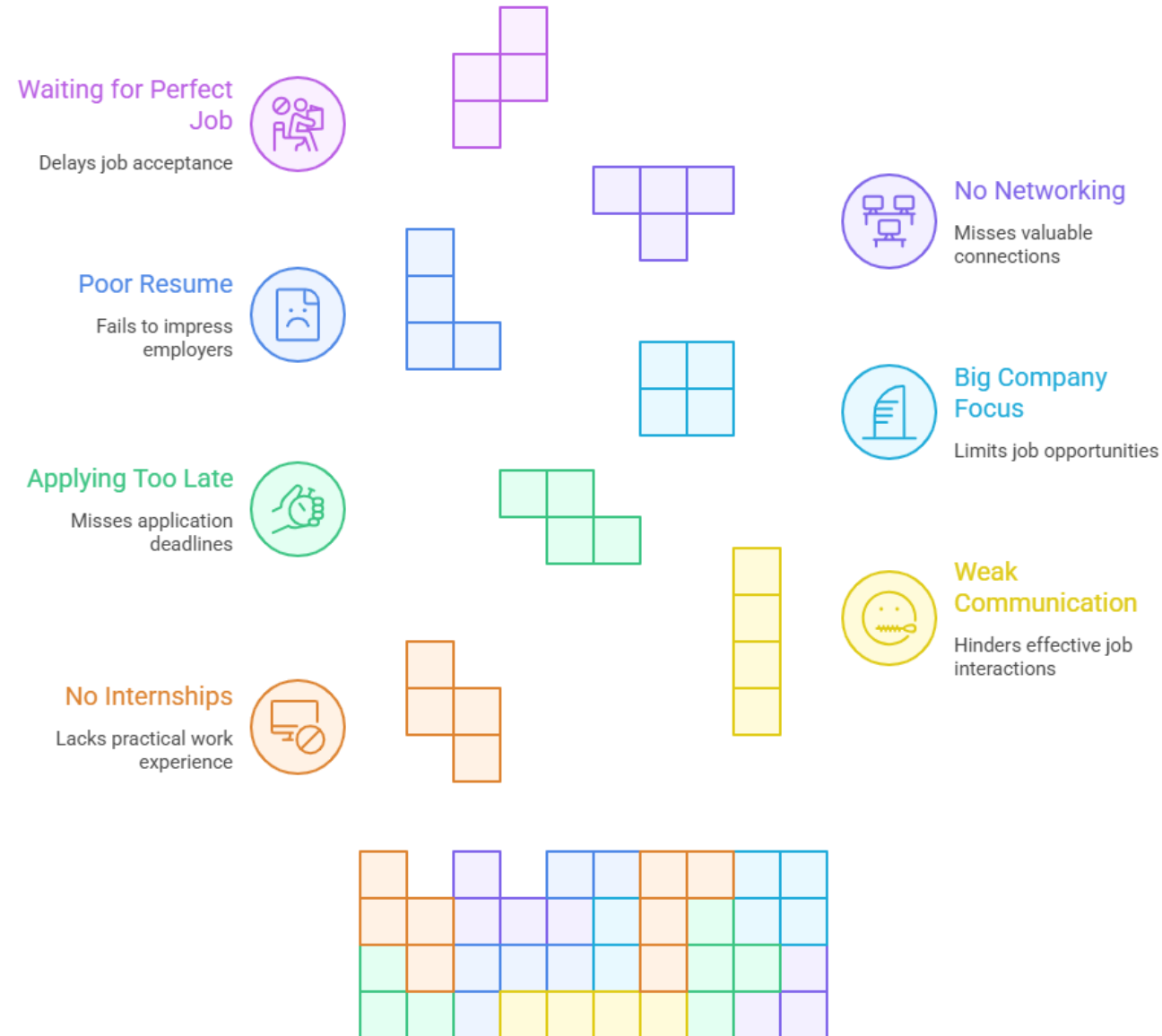
Research & Academic Pathways (Especially for PhD Graduates)

PhD holders often pursue:

- Postdoctoral Research Fellow
- Research Scientist (AI, NLP, Vision, Robotics, Data Science)
- University Lecturer / Assistant Professor
- Industry Research Labs (Applied AI, R&D teams)



International Student Job Search Hurdles



Regional Australia: Extra Benefits

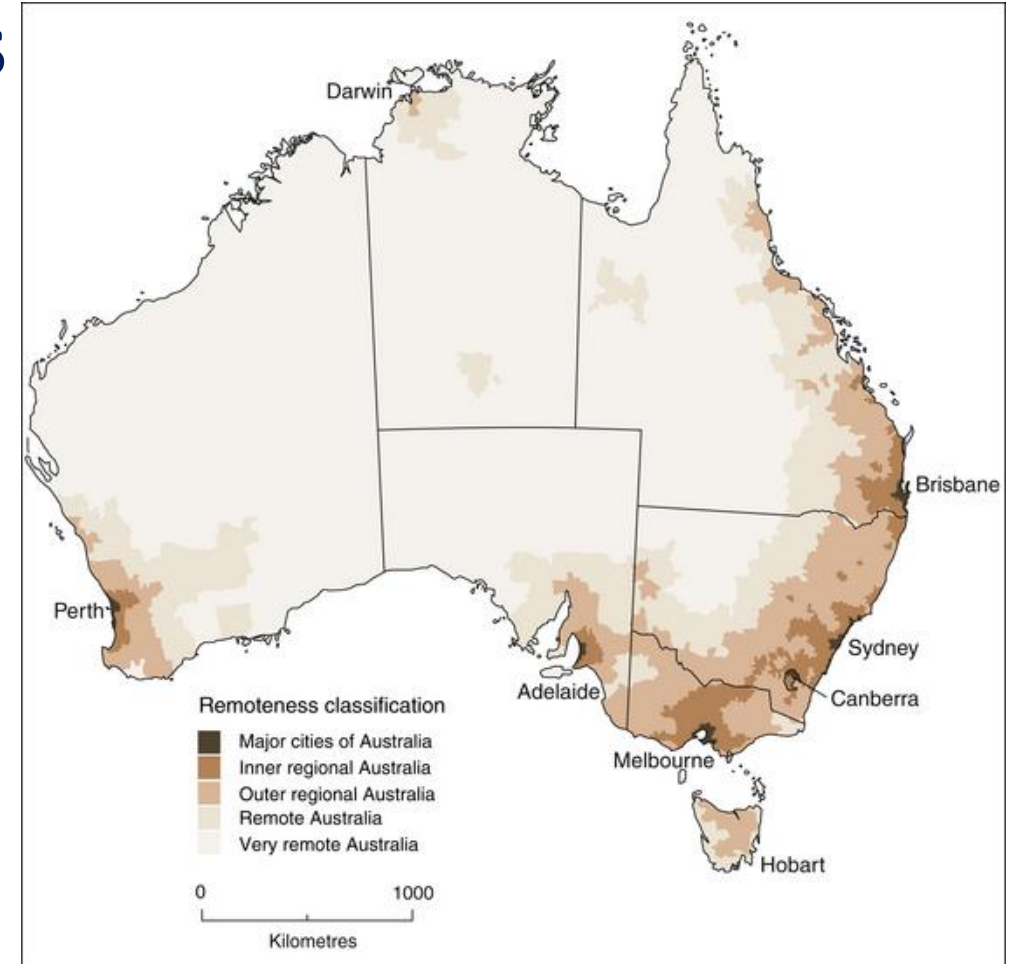
Key Regional Benefits:

- 👉 Extra migration points for PR
- 👉 Priority state nomination
- 👉 Eligibility for Subclass 491 regional visa
- 👉 Less competition for jobs
- 👉 Faster employer sponsorship in some regions
- 👉 Lower cost of living



Popular regional areas:

Adelaide, Hobart, Canberra, Wollongong, Gold Coast, Darwin, Perth



Thank you

Any Question?